



ASSIGNMENT 1

FINANCIAL ECONOMETRICS

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You can work in groups of up to three students. Write all names here. Submit your report through Canvas. You only need to submit one report per group. Submit the report in PDF format. Do not submit data or code files, only the report.

TIME SERIES LEVELS

DAILY FINANCIAL SELECT SECTOR SPDR FUND (XLF) PRICE SERIES

Note that prices and levels here for XLF are not dividend adjusted. XLF opens around \$30 in 2007 and falls to \$5 by March 2009, a decline of 83%. The recovery is slow, with the series regaining \$20 by 2015 and a level around \$30 by 2020. The COVID panic sell causes another dip to \$17, with a strong recovery through 2022 and 2023, and a sustained growth to \$50 plus by 2026. The series drifts upwards with no fixed mean, with no constant volatility, clustering around 2008 (Housing Market Crash) and 2020 (COVID).



DAILY SPDR S&P REGIONAL BANKING ETF (KRE) PRICE SERIES

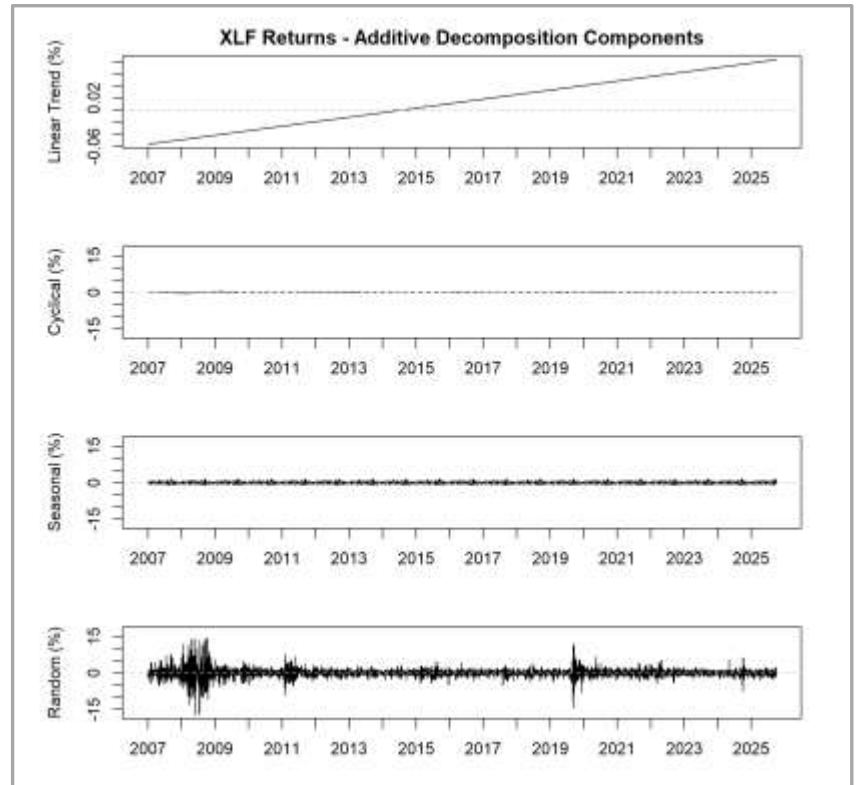
KRE opens around \$50 and bottoms at \$17 during the housing market crash with a decline of 65%. Unlike XLF, recovery is not linear. The series trades sideways between \$20 and \$30 dollars from 2010 to 2013, and climbs to high \$50s by 2018. It falls due to covid and drops to the mid \$20s, and reclaims back to \$75 by March 2023. Regional bank funding then gets pressured and it drops again to the mid \$30s before it recovers and climbs back into the \$60s and \$70s. KRE is more volatile than XLF at every stage, and the 2023 dip is specific to regional banks.



DECOMPOSITION

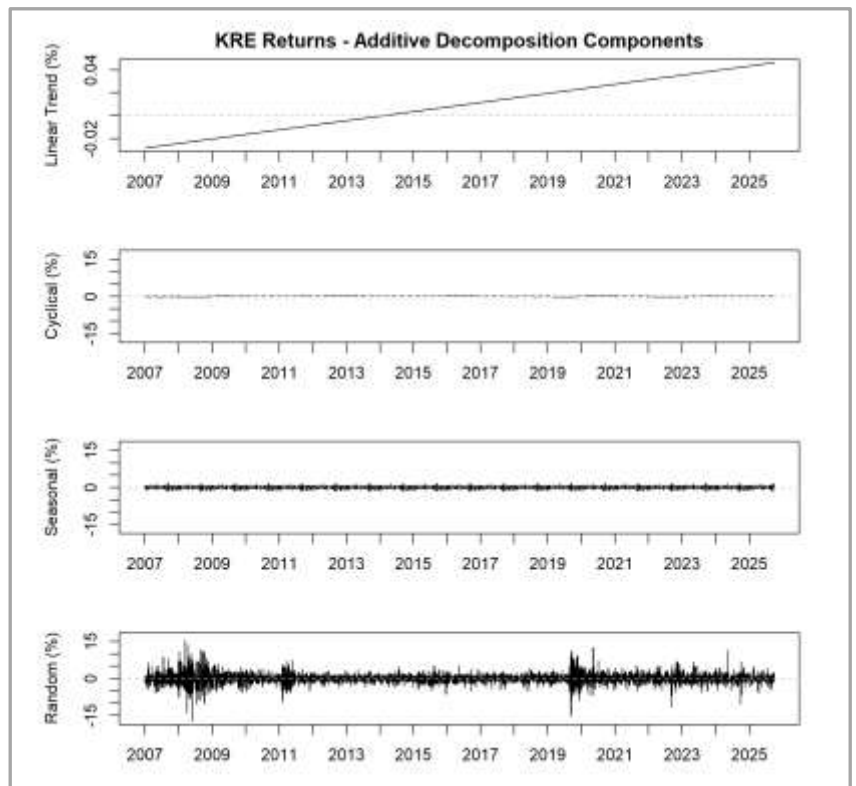
DAILY XLF RETURN SERIES DECOMPOSITION OUTPUT

The additive decomposition splits XLF into the 4 main components of trend, cyclical, seasonal, and random (noise). The result here is unambiguous. The linear trend moves from 0.06 to 0.02 across nineteen years which does immaterial on a daily return. The cyclical component sits on at about 0 and the seasonal component is negligible. The random component holds almost all the variation. This is expected since it is consistent with prices that already reflect available information. We do see that the residual is heteroskedastic, which an additive decomposition does not model.



DAILY KRE RETURN SERIES DECOMPOSITION OUTPUT

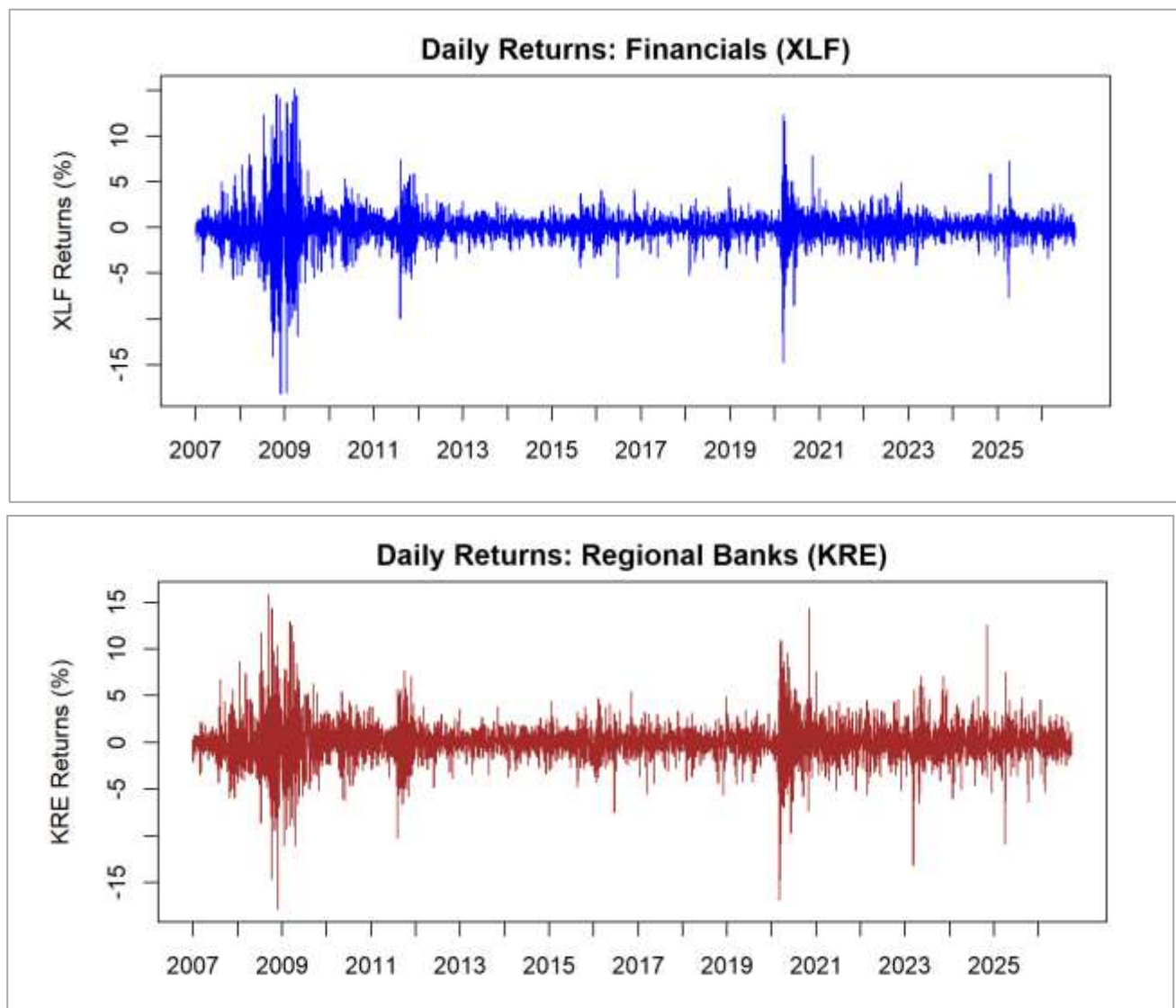
The KRE uses the same structure. The linear trend runs from 0.02 to 0.04, which is still immaterial at a daily level. The cyclical component again sits on zero and the seasonal component is negligible. As with XLF, the random noise has almost all the variance. There is a difference here because the residual panel on KRE are larger in amplitude, and show an additional cluster in 2023 that XLF does not have. Averaging across many regional banks does remove the single name risk but does not remove the risk common to the sector itself.



TIME SERIES OF THE RETURNS

DAILY XLF AND KRE ETF RETURNS

Both panels show the daily percentage returns computed as first difference of the log price. There are three main features those being: Both series fluctuate around a constant mean of about 0 with no visible drift, which is what is expected after differencing. Volatility is strongly clustered not constant, with large moves in 2008 and 2009 (housing market crash), 2020 (covid), and to a slightly less extent 2011 (end of Iraq war), and 2023 (regional bank pressure). Both series have fat tails, with XLF reaching +12% and -15% in single sessions and KRE reaching +15% and -17%. KRE is noisier, consistent with its narrower exposure, and shows a larger impact from the 2023 regional bank pressure.



STATIONARITY AND WHITE NOISE

STATIONARITY (ADF) TEST, XLF AND KRE RETURN SERIES

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AUGMENTED DICKEY-FULLER TEST: XLF RETURNS (k=0)
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Augmented Dickey-Fuller Test

data: xlf_data
Dickey-Fuller = -78.834, Lag order = 0, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(xlf_data, k = 0) : p-value smaller than printed p-value
```

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=====
AUGMENTED DICKEY-FULLER TEST: KRE RETURNS (k=0)
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Augmented Dickey-Fuller Test

data: kre_data
Dickey-Fuller = -76.435, Lag order = 0, p-value = 0.01
alternative hypothesis: stationary

Warning message:
In adf.test(kre_data, k = 0) : p-value smaller than printed p-value
```

The null hypothesis of the Augmented Dickey Fuller test is that the series is nonstationary and contains a unit root, with the alternate being that it is stationary. Both return series reject. XLF returns produce a Dickey Fuller statistic of -78.034 with a p-value of 0.01, and KRE returns -76.435 with the same reported p-value. The warning message in the outputs indicate that the true p-value is actually less than 0.01 so these are not marginal and statistical sit below any relevant critical value. I conclude that both daily return series are stationary. The price levels shows earlier would not have rejected, so taking the first difference of the logs is what gave the stationary output, which is the behavior of a difference stationary series.

WHITE NOISE TEST, XLF AND KRE RETURN SERIES

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LJUNG-BOX WHITE NOISE TEST: XLF RETURNS
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Box-Ljung test

data: xlf_data
X-squared = 221.24, df = 30, p-value < 2.2e-16
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LJUNG-BOX WHITE NOISE TEST: KRE RETURNS
=====

Box-Ljung test

data: kre_data
X-squared = 118.41, df = 30, p-value = 1.879e-12
```

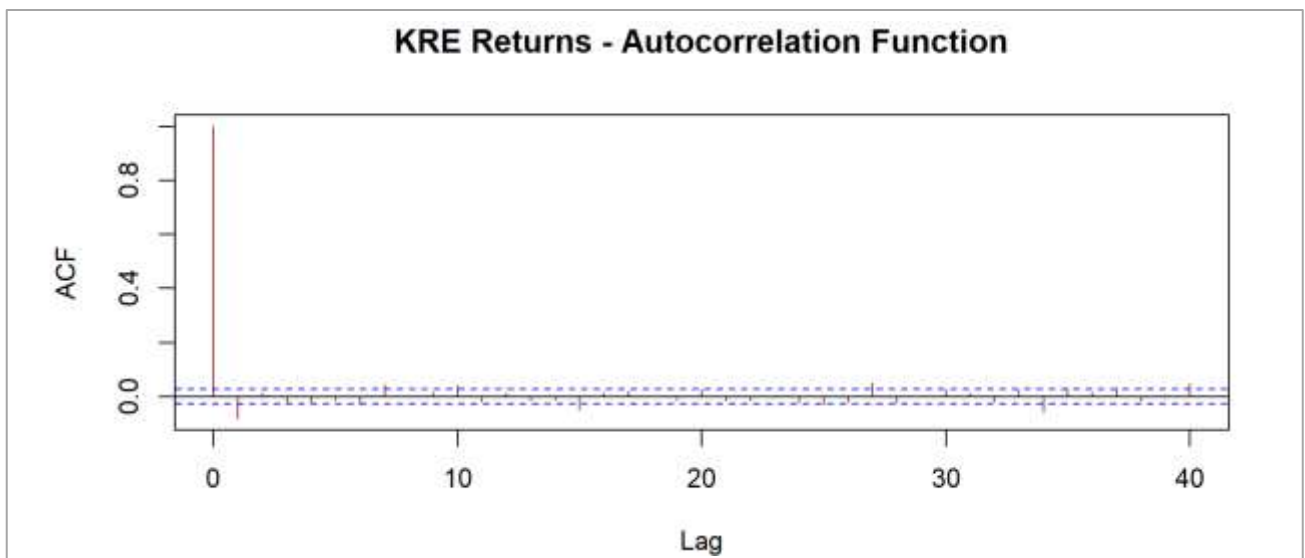
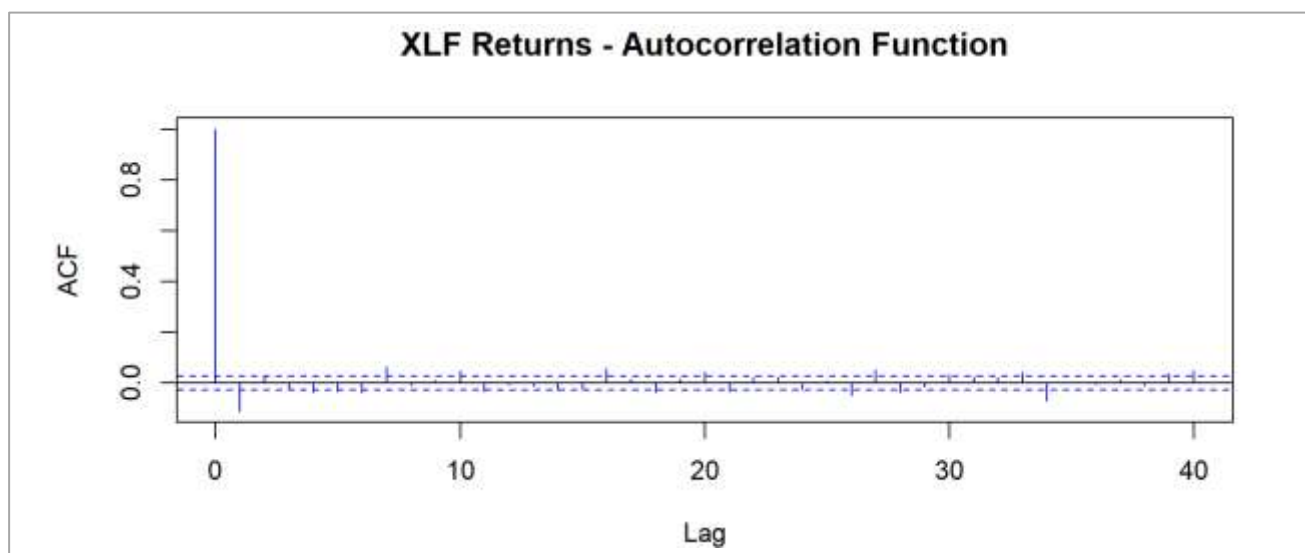
The null hypothesis of a Ljung Box test is that the series is white noise, so the autocorrelations across the tested lags are 0, and I tested 30 lags on both funds. Both reject the null. XLF returns give a chi squared statistic of 221.24 on 30 degrees of freedom with a p-value below 2.2e-16 and KRE gives 118.41 on 30 degrees of freedom with a p-value 1.879e-12.

Combining this with the ADF results, both series are stationary and not white noise, which is the condition where we can begin forecasting. The series has around 4900 observations so the test has enough power to flag autocorrelations but is not large enough to trade, and it is sensitive to the volatility clustering both series have.

ACF CHARTS

DAILY XLF AND KRE ETF RETURNS

Both correlograms show the same pattern. The spike at lag zero equals one by construction and carries no information. Alongside this, the autocorrelations collapse toward zero and stay within or close the significance bands across all 40 lags, with no gradual decay of the kind a nonstationary series would have, which confirms the ADF conclusion. A small number of bars extend marginally outside the bands in each chart, which drive the Ljung box rejections, so the two results are consistent. The important takeaway is the magnitude. The autocorrelations clearing the threshold are on the order of a few hundredths, so a model built on them would only explain the fraction of a percent of daily return variation.



DISCUSSION

DECOMPOSITION

You send your risk manager the additive decomposition results for XLF and KRE returns. She replies: "The trend component in both is close to zero, the seasonal component is negligible in both, and the residual spikes during the same crisis periods in both. I don't see any meaningful difference between these two financial-sector ETFs. Doesn't that mean we're just adding redundant exposure by holding both?" Respond to her and justify your answer.

You are reading the output correctly but drawing a incorrect conclusion. Decompsing each fund separately tells us nothing about the relationship between the two, and diversification is a question about correlation and covariance, which do not appear in these charts. The components look alike because both are equity return series dominated by irregular variation, which is true of most liquid equity series, so similarity of shape is not evidence of redundant exposure. In March 2023 KRE fell from the mid \$60s to mid \$30s on a deposit and funding stress while XLF declined much less, and the KRE residual shows a 2023 cluster that XLF does not have. The right test here is a correlation analysis focused on stress periods.

A new analyst on your team asks: "Why not just decompose the price levels of XLF and KRE directly? Prices are what we actually trade, so wouldn't that be more intuitive than decomposing returns?" Her team lead then adds: "Also, I'd argue this kind of decomposition is really only useful for a single bank stock, like JPMorgan. KRE is already a basket of regional banks, so any idiosyncratic effects have been averaged out — running a decomposition on it doesn't tell us anything new." Respond to both points and justify your reasoning.

We decompose returns because prices are nonstationary, which our ADF results confirmed for both funds. Decomposing a unit root series gives us a trend estimate that depends on almost entirely the method not the data, and the trend would absorb most of the variance by construction. Looking historically log linear trend placed the inflationary 1970s above the economic booms of the 1960s and 1990s. Returns are stationary, so their components are interpretable and comparable, and returns are the unit our risk math uses. The team leader is correct that the KRE averages away single name risk but wrong about the exercise being empty. It tells us how much variation survived that averaging and that no seasonal or cyclical structure survives either. The 2023 residual cluster is a sector risk that remains idiosyncratic with respect to XLF.

DISCUSSION

BEHAVIOR/PREDICTABILITY

A colleague argues in a meeting: "KRE diversifies away the idiosyncratic risk of any single regional bank, so what's left should mainly be systematic risk tied to the credit cycle. That means its returns should show clear mean reversion over a horizon of several months to a few years, not behave like near-random daily noise, which is what your tests seem to show." Evaluate this claim and support your conclusion with your test results.

This argument has two claims. That a basket diversifies away single name risk is reasonable, but does not follow what remains must mean revert over months to years. Exposure to the credit cycle explains why returns are systematic and says nothing about whether the price level is stationary. My test also do not settle his claim and I want to be clear about that. The ADF test rejects the unit root for KRE, daily returns, not for the KRE prices, and the price series shows the consistent drift and absent fixed mean associated with unit root. We can see this in readings with Mukherji needing data from 1926 to 2007 and a block bootstrap to find mean reversion, and found it at a 1-5 year horizon not daily data. Halkos and Kevork showed that these test have low power against slow mean reversion, so finding nothing is weak evidence.

Before you finalize your write-up, your division head pulls you aside and says: "Just so you're aware, the CIO told the board she's expecting this analysis to show that XLF and KRE returns have real, tradeable predictability. As far as she's concerned, that's the whole point of bringing in a quant analyst." Draft the actual email you would send to the CIO, and justify how you balanced honesty about your statistical findings with her expectations.

Hi,

I wanted to share the results ahead of our board discussion. Both Daily return series are stationary, with ADF statistics of -78 for XLF and -76.4 for KRE, and both reject the white noise null in Ljung Box with pvalues below $2.2e-16$ and $1.9e-12$. So there is a measureable autocorrelation in both. I want to be careful about how we present the second result. The autocorrelations on the order of a few hundredths, with about 4900 observations the test has enough power to flag effects, but not to make trades, so we cannot build a strategy on them. In summary, I found statistical structure rather than an edge. The work is useful for risk, and I would be happy to walk through it.

Best,
George Johnson